T06 (identifiability and exact ties) and T07 (size-only ranking ceiling and Bayes decisions)

SMC-BVS v3, wave 3 — proof, version 2 (after review round 1)

22 October 2026

The statement proved is `STATEMENT.md` (T06(i)–(iii), T07(i)–(iii)). Section 1 fixes notation; Section 2 proves the linear-algebra lemmas; Sections 3–5 prove T06; Section 6 proves T07. The only external input is the hypothesis H-T03 of T06(ii)(b), used in Section 4 and nowhere else. Standard facts used: the spectral theorem for real symmetric matrices and the functional calculus, the singular value decomposition, polarisation of quadratic forms, the Cauchy–Schwarz inequality and its equality case, Fubini's and Tonelli's theorems, the Gaussian density and continuity of Φ, and the definition of O_P.

1 Setting and notation

Throughout, $n \geq 3$ and $p \geq 1$. $C = I - \mathbf{1}\mathbf{1}^\top/n$ and $V = \mathrm{range}(C) = \{v \in \mathbb{R}^n : \mathbf{1}^\top v = 0\}$, so $\dim V = n - 1 \geq 2$, C is the orthogonal projection onto V, and $C\mathbf{1} = 0$. The standardised design $X \in \mathbb{R}^{n\times p}$ has columns $x_j \in V$ with $\|x_j\|^2 = n - 1$; $r_{ij} = x_i^\top x_j/(n-1)$. The contrast sphere is $\mathbb{S} = \{y \in V : \|y\|^2 = n-1\}$; it is the set of standardised responses $y = Cy^r/s_y$, $s_y^2 = \|Cy^r\|^2/(n-1)$, over non-constant raw responses y^r (if $y \in \mathbb{S}$ then $y^r = y$ gives $Cy^r = y$, $s_y = 1$). $K = \min(p, \lfloor n/2 \rfloor) \geq 1$ and $\Omega = \{S \subseteq [p] : |S| \leq K\}$. Since $n \geq 3$,

$$K \leq \lfloor n/2 \rfloor \leq n - 2 < n - 1 = \dim V. \tag{1}$$

(For $n = 3$: $1 \leq 1$; for $n \geq 4$: $n/2 \leq n - 2$.)

The score formulas below are those quoted verbatim in `STATEMENT.md` §0 from `00_COMMON.md` §1 (target: $\beta_S \sim N(0, n\sigma^2(G_S + I)^{-1})$, $\sigma^2 \sim \mathrm{IG}(0.01, 0.01)$, $\mathrm{BB}(1, p)$ capped at K). For $S \subseteq [p]$ with $k = |S|$: X_S is the $n \times k$ submatrix, $G_S = X_S^\top X_S$, $z_S = X_S^\top y$, $H_S = (n+1)G_S + I_k$, $D_S = \det H_S / \det(G_S + I_k)$, $Q_S = y^\top y - n\, z_S^\top H_S^{-1} z_S$; for $S = \emptyset$, $X_\emptyset X_\emptyset^\top := 0$, $D_\emptyset = 1$, $Q_\emptyset = y^\top y$. With $a_0 = b_0 = 0.01$ and $\nu = a_0 + n/2$,

$$m(S) = c_n\, D_S^{-1/2} (2b_0 + Q_S)^{-\nu}, \qquad c_n = \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)}.$$

The prior is $\varpi(S) = w(|S|)/W_K$ with $w(k) = \frac{1}{2}/\binom{2p-1}{k} > 0$ and $W_K = \sum_{k\leq K}\binom{p}{k} w(k)$; $F(S) = \log[\varpi(S)m(S)]$ for $S \in \Omega$, $Z = \sum_{S\in\Omega} e^{F(S)}$, $\pi(S) = e^{F(S)}/Z$ on Ω and $\pi(S) := 0$ for $S \notin \Omega$; $\mathrm{PIP}_j = \sum_{S \ni j} \pi(S)$; $\mathrm{MPM} = \{j : \mathrm{PIP}_j > \frac{1}{2}\}$. The score function of $S \in \Omega$ is $F_S : \mathbb{S} \to \mathbb{R}$, $y \mapsto F(S)$ evaluated at the response y.

The tie order $\prec$ (method §6.10) is the total order on subsets of $[p]$: smaller size first, then the lexicographically smaller sorted index list. $S^\dagger$ is the $\prec$-least element of $\arg\max_\Omega F$; for a nonempty $\mathcal{S} \subseteq \Omega$ (the scored set), $\hat{S}$ is the $\prec$-least element of $\arg\max_{\mathcal{S}} F$.

Model G: $y^r = \alpha\mathbf{1} + X^r\beta + \varepsilon$, $\varepsilon \sim N_n(0, \sigma_*^2 I)$, $\sigma_*^2 > 0$, X^r fixed, its columns being the retained (non-constant) raw columns indexed by $[p]$; D_x is the diagonal matrix of their raw sample standard deviations, which are positive, so D_x is invertible; $X = CX^r D_x^{-1}$, $\theta = D_x\beta$, $\mu = CX^r\beta = X\theta \in V$, $S_0 = \operatorname{supp}\beta$, $s_0 = |S_0|$. In the deterministic results, S_0 is an arbitrary subset of $[p]$.

A *selector* is a measurable map $\delta = \delta(y^r, U)$ into the subsets of $[p]$, where U is an auxiliary randomisation independent of y^r whose law does not depend on the truth. δ is *location-invariant* if $\delta = h(Cy^r, U)$ for a measurable h. Every measurable function of the standardised data (X, y) and U is location-invariant, since $y = Cy^r/s_y$ is a function of Cy^r; in particular $S^\dagger$, π, PIP and MPM are. (Measurability: each $F(S)$ is a continuous function of $y \in \mathbb{S}$, and $y = Cy^r/s_y$ is continuous in Cy^r off $\{Cy^r = 0\}$; so π and PIP are continuous there, and $\{S^\dagger = S\}$ and $\{\mathrm{MPM} = a\}$ are finite intersections of sets of the form $\{F(S) > F(T)\}$, $\{F(S) \geq F(T)\}$, $\{\mathrm{PIP}_j > \frac{1}{2}\}$, $\{\mathrm{PIP}_j \leq \frac{1}{2}\}$, hence Borel. On the null event $\{Cy^r = 0\}$ these maps are given the value $\emptyset$; this convention affects no probability.)

2 Preliminaries

Write $f(\lambda) = \dfrac{n\lambda}{(n+1)\lambda + 1}$ and $g(\lambda) = \dfrac{(n+1)\lambda + 1}{\lambda + 1}$ for $\lambda \geq 0$. For a real symmetric positive semidefinite $A = \sum_i \lambda_i u_i u_i^\top$ (spectral decomposition) and a function h on $[0, \infty)$, $h(A) = \sum_i h(\lambda_i) u_i u_i^\top$; this does not depend on the chosen decomposition, since $h(A) = \sum_\lambda h(\lambda) P_\lambda$ with P_λ the orthogonal projection onto the λ-eigenspace.

Lemma 2.1 (smoother and determinant)**.** *For every $S \subseteq [p]$ put $M_S = nX_S H_S^{-1} X_S^\top$ ($M_\emptyset = 0$). Then:*

(a) $M_S = f(X_S X_S^\top)$; M_S is symmetric positive semidefinite, its eigenvalues lie in $[0, \frac{n}{n+1})$, and $\operatorname{range}(M_S) = \operatorname{range}(X_S) \subseteq V$, so $\operatorname{rank} M_S \leq |S|$ and $M_S\mathbf{1} = 0$.

(b) $Q_S = y^\top(I - M_S)y$ for every $y \in \mathbb{R}^n$, and $Q_S \geq y^\top y/(n+1) \geq 0$.

(c) $D_S = \prod_{i=1}^n g(\lambda_i)$, where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $X_S X_S^\top$ (with multiplicity).

In particular M_S, D_S and, for each y, Q_S depend on S only through $X_S X_S^\top$.

Proof. For $S = \emptyset$ all claims are the conventions ($f(0) = 0$, $g(0) = 1$). Let $S \neq \emptyset$, $k = |S|$, and let $X_S = \sum_{i=1}^r s_i u_i v_i^\top$ be a thin singular value decomposition, $r = \operatorname{rank} X_S$, $s_i > 0$, $\{u_i\} \subset \mathbb{R}^n$ and $\{v_i\} \subset \mathbb{R}^k$ orthonormal; complete $\{v_i\}_{i \leq r}$ to an orthonormal basis $\{v_i\}_{i \leq k}$ of $\mathbb{R}^k$. Then $G_S = \sum_{i \leq r} s_i^2 v_i v_i^\top$, $H_S = \sum_{i \leq r}((n+1)s_i^2 + 1) v_i v_i^\top + \sum_{r < i \leq k} v_i v_i^\top$, $H_S^{-1} = \sum_{i \leq r}((n+1)s_i^2 + 1)^{-1} v_i v_i^\top + \sum_{r < i \leq k} v_i v_i^\top$, and, since $X_S v_i = s_i u_i$ for $i \leq r$ and $X_S v_i = 0$ for $i > r$,

$$M_S = nX_S H_S^{-1} X_S^\top = \sum_{i \leq r} \frac{n s_i^2}{(n+1)s_i^2 + 1} u_i u_i^\top = \sum_{i \leq r} f(s_i^2) u_i u_i^\top .$$

As $X_S X_S^\top = \sum_{i \leq r} s_i^2 u_i u_i^\top$ (eigenvalue 0 on $\{u_i\}^\perp$) and $f(0) = 0$, $M_S = f(X_S X_S^\top)$. f is strictly increasing on $[0, \infty)$ with $f(0) = 0$ and $f < \frac{n}{n+1}$, so the eigenvalues of M_S are $f(s_i^2) \in (0, \frac{n}{n+1})$ on $u_1, \ldots, u_r$ and 0 elsewhere; hence M_S is symmetric positive semidefinite, $\operatorname{range} M_S = \operatorname{span}\{u_i\}_{i \leq r} = \operatorname{range} X_S \subseteq V$ (the columns of X lie in V), $\operatorname{rank} M_S = r \leq k$, and $M_S\mathbf{1} = 0$ because $\mathbf{1} \perp V$. (b): $nz_S^\top H_S^{-1} z_S = y^\top(nX_S H_S^{-1} X_S^\top)y = y^\top M_S y$; and $y^\top(I - M_S)y \geq (1 - \frac{n}{n+1})\|y\|^2$. (c): $\det H_S = \prod_{i \leq r}((n+1)s_i^2 + 1)$ and $\det(G_S + I) = \prod_{i \leq r}(s_i^2 + 1)$, so $D_S = \prod_{i \leq r} g(s_i^2)$; the eigenvalues of $X_S X_S^\top$ are $s_1^2, \ldots, s_r^2$ and $n - r$ zeros, and $g(0) = 1$. □

Lemma 2.2 (injectivity of f on matrices)**.** *If A, B are real symmetric positive semidefinite $n \times n$ matrices with $f(A) = f(B)$, then $A = B$.*

Proof. f is continuous and strictly increasing on $[0, \infty)$, hence injective with an inverse f^{-1} on $f([0,\infty)) = [0, \frac{n}{n+1})$. Write $A = \sum_\lambda \lambda P_\lambda$ over the distinct eigenvalues of A. Then $f(A) = \sum_\lambda f(\lambda) P_\lambda$, and since the numbers $f(\lambda)$ are distinct, this is the spectral decomposition of $f(A)$: its distinct eigenvalues are the $f(\lambda)$, with eigenprojections P_λ. Hence $A = \sum_\eta f^{-1}(\eta) P_\eta^{f(A)} = f^{-1}(f(A))$, where η runs over the distinct eigenvalues of $f(A)$ and $P_\eta^{f(A)}$ are its eigenprojections. The same holds for B, so $A = f^{-1}(f(A)) = f^{-1}(f(B)) = B$. □

Lemma 2.3 (trace)**.** *For every $S \subseteq [p]$, $\operatorname{tr}(X_S X_S^\top) = \sum_{j \in S} \|x_j\|^2 = (n-1)|S|$. Hence $X_S X_S^\top = X_{S'} X_{S'}^\top$ implies $|S| = |S'|$.*

Proof. $X_S X_S^\top = \sum_{j \in S} x_j x_j^\top$ and $\operatorname{tr}(x_j x_j^\top) = \|x_j\|^2 = n - 1$; since $n - 1 > 0$, equal traces give equal sizes. □

Lemma 2.4 (scores on the sphere)**.** *Let $\kappa = 1 + 2b_0/(n-1)$ and $A_S = \kappa I - M_S$. For $S \in \Omega$ and $y \in \mathbb{S}$,*

$$F_S(y) = \log c_n + \log \varpi(S) - \tfrac{1}{2} \log D_S - \nu \log \left(y^\top A_S y \right),$$

and $v^\top A_S v \geq (\kappa - \frac{n}{n+1}) \|v\|^2 > 0$ for every $v \neq 0$. For $S, S' \in \Omega$ put $a_{S,S'} = \log \frac{\varpi(S)}{\varpi(S')} - \frac{1}{2} \log \frac{D_S}{D_{S'}}$ and $c_{S,S'} = e^{a_{S,S'}/\nu} > 0$, which depend on X only. Then for $y \in \mathbb{S}$,

$$F_S(y) = F_{S'}(y) \iff y^\top B_{S,S'}\, y = 0, \qquad B_{S,S'} := A_S - c_{S,S'} A_{S'}. \tag{2}$$

Proof. For $y \in \mathbb{S}$, $y^\top y = n-1$, so $2b_0 = \frac{2b_0}{n-1} y^\top y$ and, by Lemma 2.1(b), $2b_0 + Q_S = \kappa y^\top y - y^\top M_S y = y^\top A_S y$. Substitute into $F(S) = \log \varpi(S) + \log c_n - \frac{1}{2} \log D_S - \nu \log(2b_0 + Q_S)$. Positivity: the eigenvalues of M_S are $< \frac{n}{n+1} < 1 < \kappa$. Finally $F_S(y) - F_{S'}(y) = a_{S,S'} - \nu \log \frac{y^\top A_S y}{y^\top A_{S'} y}$, which vanishes iff $y^\top A_S y = c_{S,S'}\, y^\top A_{S'} y$. □

Lemma 2.5 (quadratic forms vanishing on V)**.** *Let B be a real symmetric $n \times n$ matrix. If $v^\top B v = 0$ for all $v \in \mathbb{S}$, then $v^\top B v = 0$ for all $v \in V$, and $CBC = 0$.*

Proof. For $v \in V \setminus \{0\}$, $y = \sqrt{n-1}\, v/\|v\| \in \mathbb{S}$ and $v^\top B v = \frac{\|v\|^2}{n-1} y^\top B y = 0$; also $0^\top B 0 = 0$. For $u, v \in V$, polarisation gives $u^\top B v = \frac{1}{4}[(u+v)^\top B (u+v) - (u-v)^\top B (u-v)] = 0$. Applying this to $u = Ca$, $v = Cb$ for arbitrary $a, b \in \mathbb{R}^n$ gives $a^\top CBCb = 0$, so $CBC = 0$. □

Lemma 2.6 (null sets of quadratic forms)**.** *Let $d \geq 1$ and let $q(t) = t^\top B t$ with B a real symmetric $d \times d$ matrix, $B \neq 0$. Then $\{t \in \mathbb{R}^d : q(t) = 0\}$ has Lebesgue measure 0. Consequently, if T is a random vector in $\mathbb{R}^d$ with a density, $\Pr(q(T) = 0) = 0$.*

Proof. Write $B = O \Lambda O^\top$ with O orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$; since $B \neq 0$, some $\lambda_i \neq 0$, say λ_1. The orthogonal map $t \mapsto O^\top t$ preserves Lebesgue measure, so it suffices to show that $N = \{u : \sum_i \lambda_i u_i^2 = 0\}$ is null. For each fixed $(u_2, \ldots, u_d)$ the section $\{u_1 : \lambda_1 u_1^2 = -\sum_{i \geq 2} \lambda_i u_i^2\}$ has at most two points, hence one-dimensional measure 0; N is closed, hence measurable, and by Fubini's theorem its d-dimensional measure is 0. The last claim holds because a law with a density gives probability 0 to Lebesgue-null sets. □

3 T06(i): exact ties

Theorem 3.1 (T06(i)(a)). *Let $S, S' \in \Omega$. Then $F_S = F_{S'}$ on $\mathbb{S}$ if and only if $X_S X_S^\top = X_{S'} X_{S'}^\top$. If $X_S X_S^\top = X_{S'} X_{S'}^\top$, then $|S| = |S'|$ and $F(S) = F(S')$ for every response vector $y \in \mathbb{R}^n$ put into the formulas of Section 1.*

Proof. If. Suppose $X_S X_S^\top = X_{S'} X_{S'}^\top$. By Lemma 2.3, $|S| = |S'|$, so $\varpi(S) = \varpi(S')$. By Lemma 2.1, $M_S = M_{S'}$ and $D_S = D_{S'}$, and for every $y \in \mathbb{R}^n$, $Q_S = y^\top (I - M_S) y = Q_{S'}$. Hence $m(S) = m(S')$ and $F(S) = F(S')$ for every $y \in \mathbb{R}^n$, in particular on $\mathbb{S}$.

Only if. Suppose $F_S = F_{S'}$ on $\mathbb{S}$. Write $c = c_{S,S'} > 0$. By (2), $y^\top (A_S - c A_{S'}) y = 0$ for all $y \in \mathbb{S}$, so by Lemma 2.5, $C(A_S - cA_{S'})C = 0$. By Lemma 2.1(a), M_S and $M_{S'}$ are symmetric with range in V, so $CM_SC = M_S$ and $CM_{S'}C = M_{S'}$; also $CIC = C$. Therefore

$$(1-c)\kappa\, C = M_S - cM_{S'}. \tag{3}$$

By Lemma 2.1(a) and (1), $\operatorname{rank} M_S \le |S| \le K < n - 1 = \dim V$. M_S maps V into V, so its restriction to V has a nontrivial kernel: there is $u \in V$, $u \neq 0$, with $M_S u = 0$. Applying (3) to u and taking the inner product with u (note $Cu = u$):

$$(1-c)\kappa \|u\|^2 = -c\, u^\top M_{S'} u \le 0,$$

because $M_{S'} \succeq 0$ and $c > 0$; as $\kappa\|u\|^2 > 0$, $c \ge 1$. In the same way there is $u' \in V \setminus \{0\}$ with $M_{S'} u' = 0$, and $(1-c)\kappa\|u'\|^2 = u'^\top M_S u' \ge 0$ gives $c \le 1$. Hence $c = 1$, and (3) gives $M_S = M_{S'}$, i.e. $f(X_S X_S^\top) = f(X_{S'} X_{S'}^\top)$ by Lemma 2.1(a). Both matrices are symmetric positive semidefinite, so Lemma 2.2 gives $X_S X_S^\top = X_{S'} X_{S'}^\top$. □

Example 3.2 (a valid design for every $n \ge 4$). Let $n \ge 4$, $p = 4$, $c = \sqrt{n-1}/2$ and

$$x_1 = c\,(1,1,-1,-1,0,\dots,0)^\top, \quad x_2 = c\,(1,-1,1,-1,0,\dots,0)^\top, \quad x_3 = \tfrac{1}{\sqrt{2}}(x_1+x_2), \quad x_4 = \tfrac{1}{\sqrt{2}}(x_1-x_2),$$

vectors in $\mathbb{R}^n$ (zeros after the fourth coordinate). Then $x_1^\top x_2 = c^2(1-1-1+1) = 0$, $\|x_1\|^2 = \|x_2\|^2 = 4c^2 = n-1$ and $\|x_1 \pm x_2\|^2 = 2(n-1)$, so every x_j has entries summing to 0 and $\|x_j\|^2 = n-1$. Thus $X = [x_1, \dots, x_4]$ is a standardised design, arising from the valid raw input $X^r = X$ together with any non-constant raw response y^r (all entries finite, $n \ge 3$, no constant column; the method's standardisation returns X itself, since each column is centred with sample SD 1). $K = \min(4, \lfloor n/2 \rfloor) \ge 2$; all models below have size $\le 2 \le K$. (For $n = 5$: $x_1 = (1,1,-1,-1,0)^\top$, $x_2 = (1,-1,1,-1,0)^\top$.) The correlations are computed in the proof of Proposition 5.4(b).

Corollary 3.3 (T06(i)(b)). *If $S, S' \in \Omega$ and $X_S X_S^\top \neq X_{S'} X_{S'}^\top$ (in particular if* $\operatorname{span} X_S = \operatorname{span} X_{S'}$ *but the outer products differ), then $F_S(y) \neq F_{S'}(y)$ for some $y \in \mathbb{S}$. Such pairs exist: in Example 3.2, $S = \{1,2\}$ and $S' = \{1,3\}$ are feasible,* $\operatorname{span} X_S = \operatorname{span} X_{S'} = \operatorname{span}\{x_1, x_2\}$, *and $X_S X_S^\top \neq X_{S'} X_{S'}^\top$.*

Proof. The first claim is the contrapositive of the "only if" direction of Theorem 3.1. In the example, $x_3 \in \operatorname{span}\{x_1, x_2\}$ and $x_2 = \sqrt{2} x_3 - x_1$, so the spans agree. Applying both outer products to x_2: $X_S X_S^\top x_2 = x_1(x_1^\top x_2) + x_2(x_2^\top x_2) = (n-1)x_2$, while $X_{S'} X_{S'}^\top x_2 = x_1(x_1^\top x_2) + x_3(x_3^\top x_2) = \frac{n-1}{\sqrt{2}} x_3 = \frac{n-1}{2}(x_1 + x_2)$, and $(n-1)x_2 \neq \frac{n-1}{2}(x_1+x_2)$ since $x_1 \neq x_2$. □

Lemma 3.4 (the case $K = 1$). *Suppose $K = 1$ (for instance $n = 3$, or $p = 1$). Then (a) no two distinct $S, S' \in \Omega$ have* $\operatorname{span} X_S = \operatorname{span} X_{S'}$ *and $X_S X_S^\top \neq X_{S'} X_{S'}^\top$; (b) if $S \neq S'$ in Ω have $X_S X_S^\top = X_{S'} X_{S'}^\top$, then $\{S, S'\} = \{\{i\}, \{j\}\}$ with $i \neq j$ and $|r_{ij}| = 1$. Hence the existence claims of Corollary 3.3 and Proposition 5.4(b) need $K \ge 2$, and by Example 3.2 they hold for some valid input at every $n \ge 4$.*

Proof. $\Omega = \{\emptyset\} \cup \{\{j\} : j \in [p]\}$, and every $x_j \neq 0$ (its squared norm is $n-1 > 0$). (a) $\operatorname{span} X_\emptyset = \{0\} \neq \operatorname{span}\{x_j\}$, so equal spans with $S \neq S'$ force $S = \{i\}$, $S' = \{j\}$, $i \neq j$, and $x_i = \lambda x_j$; equal norms give $\lambda = \pm 1$, so $x_i x_i^\top = x_j x_j^\top$. (b) By Lemma 2.3, $|S| = |S'|$; the only model of size 0 is $\emptyset$, so $S = \{i\}$, $S' = \{j\}$ with $i \neq j$, and $x_i x_i^\top = x_j x_j^\top$; applying both sides to x_i gives $(n-1)x_i = (n-1)r_{ij}x_j$, and taking norms, $|r_{ij}| = 1$. □

Theorem 3.5 (T06(i)(c))**.** *Assume G. Then, P_0-almost surely, $Cy^r \neq 0$ and every pair $S \neq S'$ in Ω with $F(S) = F(S')$ at the observed y satisfies $X_S X_S^\top = X_{S'} X_{S'}^\top$. Consequently, under G and NIS, $P_0(\exists\, S \neq S' \in \Omega :\ F(S) = F(S')) = 0$.*

Proof. Let $U \in \mathbb{R}^{n \times (n-1)}$ have orthonormal columns spanning V, so $U^\top U = I_{n-1}$ and $UU^\top = C$. Under G, $v := Cy^r = C(\alpha \mathbf{1} + X^r\beta) + C\varepsilon = \mu + C\varepsilon$, and its coordinates $t := U^\top v = U^\top \mu + U^\top \varepsilon$ (as $U^\top C = U^\top U U^\top = U^\top$) satisfy $t \sim N_{n-1}(U^\top \mu, \sigma_*^2 I_{n-1})$, a law with a density on $\mathbb{R}^{n-1}$; and $v = Ut$. In particular $P_0(v = 0) = P_0(t = 0) = 0$. On $\{v \neq 0\}$, $y = v/s_y$ with $s_y = \|v\|/\sqrt{n-1} > 0$, and $y \in \mathbb{S}$.

Fix $S \neq S'$ in Ω with $X_S X_S^\top \neq X_{S'} X_{S'}^\top$, and let $B = B_{S,S'}$ of Lemma 2.4. By (2) and homogeneity (y is a positive multiple of v), on $\{v \neq 0\}$: $F(S) = F(S') \iff v^\top B v = 0 \iff t^\top (U^\top B U) t = 0$. The matrix $U^\top B U$ is not 0: otherwise $v^\top B v = t^\top U^\top B U t = 0$ for every $v = Ut \in V$, hence $F_S = F_{S'}$ on $\mathbb{S}$ by (2), and Theorem 3.1 would give $X_S X_S^\top = X_{S'} X_{S'}^\top$. By Lemma 2.6 with $d = n-1$, $P_0(v \neq 0,\ F(S) = F(S')) \leq P_0(t^\top U^\top B U t = 0) = 0$. There are finitely many pairs, so, P_0-a.s., no pair with unequal outer products ties. Under NIS every pair $S \neq S'$ in Ω has unequal outer products, which gives the last claim. □

Remark 3.6. The argument uses only that Cy^r has a density on V; no condition on μ (hence on β, S_0, or the signal strength) is needed.

4 T06(ii): the fallback without NIS

Proposition 4.1 (T06(ii)(a))**.** *For every valid input, every $S_0 \subseteq [p]$ and every response: if $\pi(S_0) > \frac{1}{2}$, then $S_0 \in \Omega$ and $F(S_0) > F(S)$ for every $S \in \Omega \setminus \{S_0\}$. Hence $\arg\max_\Omega F = \{S_0\}$ and $S^\dagger = S_0$ whatever the tie rule.*

Proof. $\pi(S) = 0$ for $S \notin \Omega$, so $\pi(S_0) > 0$ forces $S_0 \in \Omega$. For $S \in \Omega \setminus \{S_0\}$, $\pi(S) \leq \sum_{T \neq S_0} \pi(T) = 1 - \pi(S_0) < \frac{1}{2} < \pi(S_0)$. Since $F = \log \pi + \log Z$ on Ω and log is strictly increasing, $F(S) < F(S_0)$. So S_0 is the unique maximiser and any tie rule returns it. □

Proposition 4.2 (T06(ii)(b))**.** *Assume H-T03: the conclusion of T03(i) holds along the sequence considered. (The conjecture file `theorems/T03_A2_selection_consistency.md` states, verbatim: "In R1 under G, R, SE and BM-C (or BM-∞): (i) $1 - \pi(S_0) = O_{P_0}(r_n)$", with $r_n = \sqrt{\log n / n}$. T03 is unproved; this proposition is conditional on it.) Then $P_0(\pi(S_0) > \frac{1}{2}) \to 1$; on this event the conclusions of Proposition 4.1 hold. NIS is not used.*

Proof. Let $\epsilon > 0$. By the definition of O_{P_0}, there are $M < \infty$ and n_1 with $P_0(1 - \pi(S_0) > M r_n) < \epsilon$ for $n \geq n_1$. Since $r_n \to 0$, $M r_n < \frac{1}{2}$ for $n \geq n_2$. For $n \geq \max(n_1, n_2)$, $P_0(\pi(S_0) \leq \frac{1}{2}) = P_0(1 - \pi(S_0) \geq \frac{1}{2}) \leq P_0(1 - \pi(S_0) > M r_n) < \epsilon$. □

Remark 4.3. Only the consequence $\pi(S_0) \to 1$ in P_0-probability of H-T03 is used. If $S_0 \in \Omega$ had a "twin" $S' \neq S_0$ in Ω with $X_{S'} X_{S'}^\top = X_{S_0} X_{S_0}^\top$, then $\pi(S') = \pi(S_0)$ by Theorem 3.1 and so $\pi(S_0) \leq \frac{1}{2}$ for every response; thus H-T03 forces S_0 to have no twin in Ω for all large n, without any assumption of NIS for the other models.

5 T06(iii): duplicates and weak separation

Assume DUP(a): $i \neq j$ and $x_i = s\,x_j$ with $s \in \{-1, 1\}$ (by Cauchy–Schwarz and $\|x_i\| = \|x_j\|$, this is equivalent to $|r_{ij}| = 1$). Let τ be the transposition of i and j on $[p]$, extended to subsets by $\tau S = \{\tau k : k \in S\}$.

Proposition 5.1 (T06(iii)(a)(b))**.** *(a) For every $S \in \Omega$: $|\tau S| = |S|$, $\tau S \in \Omega$, $X_{\tau S} X_{\tau S}^\top = X_S X_S^\top$, and $F(\tau S) = F(S)$ for every response. (b) If $S \in \Omega$ and $|S \cap \{i, j\}| = 1$, then $\tau S \neq S$, $\pi(\tau S) = \pi(S)$ and $\pi(S) < \frac{1}{2}$. Moreover* $\mathrm{PIP}_i = \mathrm{PIP}_j$.

Proof. (a) τ is a bijection of $[p]$, so $|\tau S| = |S| \leq K$. For every $k \in [p]$, $x_{\tau k} x_{\tau k}^\top = x_k x_k^\top$: trivially if $k \notin \{i, j\}$, and $x_i x_i^\top = s^2 x_j x_j^\top = x_j x_j^\top$. Hence $X_{\tau S} X_{\tau S}^\top = \sum_{k \in S} x_{\tau k} x_{\tau k}^\top = \sum_{k \in S} x_k x_k^\top = X_S X_S^\top$, and Theorem 3.1 ("if" part) gives $F(\tau S) = F(S)$ for every response. (b) If S contains exactly one of i, j, then τS contains the other and not the first, so $\tau S \neq S$; by (a), $\pi(\tau S) = \pi(S)$. Both are nonempty, so $\emptyset \notin \{S, \tau S\}$, and $\emptyset \in \Omega$ has $\pi(\emptyset) > 0$. Hence $2\pi(S) = \pi(S) + \pi(\tau S) \leq 1 - \pi(\emptyset) < 1$. For the PIPs, $S \mapsto \tau S$ is a bijection of Ω mapping $\{S \ni i\}$ onto $\{S \ni j\}$, so $\mathrm{PIP}_j = \sum_{S \ni i} \pi(\tau S) = \sum_{S \ni i} \pi(S) = \mathrm{PIP}_i$. □

Theorem 5.2 (T06(iii)(c), two-point bound)**.** *Let $\theta_k = (\alpha_k, \beta_k)$, $k = 1, 2$, be two truths under G with the same X^r and the same $\sigma_*^2 > 0$, with supports $S_1 \neq S_2$; $\mu_k = C X^r \beta_k$, $m_k = \alpha_k \mathbf{1} + X^r \beta_k$. Let $\delta = \delta(y^r, U)$ be a selector (U independent of y^r, with a law λ not depending on k). Then*

$$\max_k P_{\theta_k}(\delta \neq S_k) \;\geq\; \tfrac{1}{2} \sum_k P_{\theta_k}(\delta \neq S_k) \;\geq\; \Phi\Big(- \frac{\|m_1 - m_2\|}{2\sigma_*} \Big), \tag{4}$$

and $\|m_1 - m_2\|^2 = \|\mu_1 - \mu_2\|^2 + \frac{1}{n}\big(\mathbf{1}^\top (m_1 - m_2)\big)^2$; so $\|m_1 - m_2\| = \|\mu_1 - \mu_2\|$ iff the pair is mean-matched*, $\mathbf{1}^\top m_1 = \mathbf{1}^\top m_2$, i.e. $\alpha_1 + \bar{x}^{r\top} \beta_1 = \alpha_2 + \bar{x}^{r\top} \beta_2$ with $\bar{x}^r = X^{r\top} \mathbf{1}/n$ (this is not $\alpha_1 = \alpha_2$). If δ is location-invariant, then $\max_k P_{\theta_k}(\delta \neq S_k) \geq \Phi(-\|\mu_1 - \mu_2\|/(2\sigma_*))$ for any intercepts.*

Proof. Let p_k be the $N_n(m_k, \sigma_*^2 I)$ density of y^r under θ_k and $\varphi(y^r) = \lambda\{u : \delta(y^r, u) = S_1\} \in [0, 1]$, measurable by Fubini's theorem. Since $S_1 \neq S_2$, $\{\delta = S_1\} \subseteq \{\delta \neq S_2\}$, and by independence of U,

$$P_{\theta_1}(\delta \neq S_1) + P_{\theta_2}(\delta \neq S_2) \geq \int (1 - \varphi) p_1 + \int \varphi p_2 \geq \int \min(p_1, p_2),$$

the last step pointwise since $(1 - \varphi)p_1 + \varphi p_2$ is a convex combination of p_1, p_2. Let $\Delta = \|m_1 - m_2\|$. If $\Delta = 0$, $p_1 = p_2$ and $\int \min(p_1, p_2) = 1 = 2\Phi(0)$. If $\Delta > 0$, let $e = (m_2 - m_1)/\Delta$ and $T(y) = e^\top (y - \frac{1}{2}(m_1 + m_2))$. Then $\|y - m_1\|^2 - \|y - m_2\|^2 = 2(m_2 - m_1)^\top y + \|m_1\|^2 - \|m_2\|^2 = 2\Delta T(y)$, so $p_2(y) > p_1(y) \iff T(y) > 0$, and

$$\int \min(p_1, p_2) = P_{\theta_1}(T > 0) + P_{\theta_2}(T \leq 0).$$

Under θ_1, $T \sim N(e^\top (m_1 - m_2)/2, \sigma_*^2) = N(-\Delta/2, \sigma_*^2)$, so $P_{\theta_1}(T > 0) = \Phi(-\Delta/(2\sigma_*))$; under θ_2, $T \sim N(\Delta/2, \sigma_*^2)$ and $P_{\theta_2}(T \leq 0) = \Phi(-\Delta/(2\sigma_*))$. Dividing by 2 and using max $\geq$ average gives (4).

The decomposition: $m_1 - m_2 = C(m_1 - m_2) + (I - C)(m_1 - m_2)$ is orthogonal, $C(m_1 - m_2) = C X^r (\beta_1 - \beta_2) = \mu_1 - \mu_2$ as $C\mathbf{1} = 0$, and $(I - C)(m_1 - m_2) = \mathbf{1}\,\mathbf{1}^\top (m_1 - m_2)/n$ has squared norm $(\mathbf{1}^\top (m_1 - m_2))^2/n$.

Location-invariant case: $\delta = h(Cy^r, U)$. Under θ_k, $Cy^r = \mu_k + C\varepsilon$, whose law depends on θ_k only through μ_k; so $P_{\theta_k}(\delta \neq S_k)$ is unchanged if α_2 is replaced by $\alpha_2' = \alpha_2 + \mathbf{1}^\top (m_1 - m_2)/n$. With $m_2' = \alpha_2' \mathbf{1} + X^r \beta_2$, $\mathbf{1}^\top (m_1 - m_2') = \mathbf{1}^\top (m_1 - m_2) - n \cdot \mathbf{1}^\top (m_1 - m_2)/n = 0$, so $\|m_1 - m_2'\| = \|\mu_1 - \mu_2\|$, and (4) for the pair $(\theta_1, (\alpha_2', \beta_2))$ gives the claim. □

Corollary 5.3 (T06(iii)(d), no uniform recovery)**.** *For each n let Θ_n be a class of truths under G containing a pair $\theta_{1,n}, \theta_{2,n}$ with a common design X_n^r, distinct supports, a common noise variance σ_n^2, and $\|\mu_{1,n} - \mu_{2,n}\|/\sigma_n \to 0$. Then every sequence of location-invariant selectors δ_n satisfies $\liminf_n \sup_{\theta\in\Theta_n} P_\theta(\delta_n \neq S_0(\theta)) \geq \frac{1}{2}$; the same holds for every sequence of selectors when the pairs are mean-matched ($\mathbf{1}^\top m_{1,n} = \mathbf{1}^\top m_{2,n}$). In particular $\sup_{\Theta_n} P_\theta(\delta_n \neq S_0(\theta)) \not\to 0$. Under DUP(a) with $x_i = s\,x_j$, the truths $\theta^* = b\,e_i$ and $\theta^* = s\,b\,e_j$ ($b \neq 0$, $\beta = D_x^{-1}\theta^*$) (with the same X^r and the same σ_*^2, intercepts arbitrary) have supports $\{i\} \neq \{j\}$ and $\mu_1 = \mu_2$, so every location-invariant selector has maximal error $\geq \frac{1}{2}$ over this pair.*

Proof. $\sup_{\Theta_n} P_\theta(\delta_n \neq S_0(\theta)) \geq \max_k P_{\theta_{k,n}}(\delta_n \neq S_{k,n}) \geq \Phi(-\|\mu_{1,n} - \mu_{2,n}\|/(2\sigma_n)) \to \Phi(0) = \frac{1}{2}$ by Theorem 5.2 (with $\|m_{1,n} - m_{2,n}\| = \|\mu_{1,n} - \mu_{2,n}\|$ in the mean-matched case) and continuity of Φ. For the duplicate pair, $\mu = X\theta^*$ (since $X\theta^* = CX^r D_x^{-1} D_x \beta$), so $\mu_1 = b\,x_i$ and $\mu_2 = s\,b\,x_j = s^2 b\,x_i = b\,x_i$ (as $x_j = s\,x_i$); supports of β equal those of θ^* since D_x is diagonal and positive. Theorem 5.2 with $\|\mu_1 - \mu_2\| = 0$ gives the bound $\Phi(0) = \frac{1}{2}$. □

Proposition 5.4 (T06(iii)(e), the near-duplicate report)**.** *(a) Let $i \neq j$, $A \subseteq [p]\backslash\{i,j\}$, $|A|+1 \leq K$. Then $F_{A\cup\{i\}} = F_{A\cup\{j\}}$ on $\mathbb{S}$ if and only if $|r_{ij}| = 1$. So a reported pair ($|r_{ij}| \geq 1 - 10^{-10}$) with $|r_{ij}| < 1$ gives no such identity. (b) For every $n \geq 4$, in Example 3.2, $S = \{1,2\}$ and $S' = \{3,4\}$ are distinct feasible models with $X_S X_S^\top = X_{S'} X_{S'}^\top$ (identical score functions by Theorem 3.1), while $\max_{i\neq j} |r_{ij}| = 1/\sqrt{2} < 1 - 10^{-10}$, so the method reports no pair.*

Proof. (a) Both models lie in Ω and are distinct. By Theorem 3.1, the score functions agree iff $X_A X_A^\top + x_i x_i^\top = X_A X_A^\top + x_j x_j^\top$, i.e. iff $x_i x_i^\top = x_j x_j^\top$. If this holds, applying both sides to x_i gives $(n-1)x_i = (x_j^\top x_i)x_j = (n-1)r_{ij}x_j$, so $x_i = r_{ij}x_j$, and taking norms, $|r_{ij}| = 1$. Conversely, $|r_{ij}| = 1$ is equality in Cauchy–Schwarz, so $x_i = \pm x_j$ and $x_i x_i^\top = x_j x_j^\top$. (b) $x_3 x_3^\top + x_4 x_4^\top = \frac{1}{2}[(x_1+x_2)(x_1+x_2)^\top + (x_1-x_2)(x_1-x_2)^\top] = x_1 x_1^\top + x_2 x_2^\top$. Both models have size $2 \leq K$. The correlations: $r_{12} = 0$; $r_{13} = x_1^\top(x_1+x_2)/(\sqrt{2}(n-1)) = 1/\sqrt{2}$, and likewise $r_{14} = r_{23} = 1/\sqrt{2}$, $r_{24} = -1/\sqrt{2}$; $r_{34} = \frac{\|x_1\|^2 - \|x_2\|^2}{2(n-1)} = 0$. So $\max_{i\neq j} |r_{ij}| = 1/\sqrt{2} < 1 - 10^{-10}$. When $K = 1$, Lemma 3.4(b) shows that every such identity is between a reported pair. □

Remark 5.5. Proposition 5.4 shows that the report is neither sufficient nor necessary for an exact score identity: within a reported component, pairs with $|r_{ij}| < 1$ never give identities of the form in (a) (and, under G, tie with probability 0 unless the outer products are equal, by Theorem 3.5), while identities can occur between models that contain no reported pair. The report is therefore a warning about near-collinearity, not a statement of exact (non-)identifiability.

6 T07: size-only ceiling and Bayes decisions

Remark 6.1 (the tie order). The arguments of this section use only that $\prec$ is a total order on subsets of $[p]$, one and the same for $S_{\tilde{w}}^\dagger$, $S^\dagger$, S_k^* and $\hat{S}$, and the fact that the $\prec$-least element of a finite set is also $\prec$-least in every subset containing it. So all results hold verbatim for any fixed total order used throughout; the method's order is one instance.

Throughout this section m enters only through $m(S) > 0$ for every $S \subseteq [p]$ (Lemma 2.1(b) gives $2b_0 + Q_S > 0$, and $D_S > 0$). For a positive size-only prior on Ω, $\tilde{w}(k) > 0$ ($0 \leq k \leq K$), $\varpi_{\tilde{w}}(S) = \tilde{w}(|S|)/\tilde{W}$ with $\tilde{W} = \sum_{k\leq K} \binom{p}{k} \tilde{w}(k)$; $S_{\tilde{w}}^\dagger$ is the $\prec$-least element of $\arg\max_\Omega \varpi_{\tilde{w}} m$; S_k^* ($0 \leq k \leq p$) is the $\prec$-least element of $\arg\max_{|S|=k} m$, well defined since $\{|S| = k\}$ is finite and nonempty.

Theorem 6.2 (T07(i), the ceiling)**.** *For every response and every positive size-only prior $\tilde{w}$: if $S^\dagger_{\tilde{w}} = S_0$, then $s_0 \le K$ and $S^*_{s_0} = S_0$. Equivalently, $\{S^\dagger_{\tilde{w}} = S_0\} \subseteq \{S^*_{s_0} = S_0\}$ (the left event being empty if $s_0 > K$), and on $\{S^*_{s_0} \neq S_0\}$, $S^\dagger_{\tilde{w}} \neq S_0$ for every positive size-only prior.*

Proof. Suppose $S^\dagger_{\tilde{w}} = S_0$. Then $S_0 \in \Omega$, so $s_0 \le K$, and $\mathcal{K} = \{S : |S| = s_0\} \subseteq \Omega$. Let $M = \max_\Omega \varpi_{\tilde{w}} m = \varpi_{\tilde{w}}(S_0)m(S_0)$. On $\mathcal{K}$, $\varpi_{\tilde{w}}(S)m(S) = \gamma\, m(S)$ with $\gamma = \tilde{w}(s_0)/\tilde{W} > 0$. For $S \in \mathcal{K}$, $\gamma m(S) \le M = \gamma m(S_0)$, so $S_0 \in \mathcal{A} := \arg\max_{\mathcal{K}} m$, and every $S \in \mathcal{A}$ has $\varpi_{\tilde{w}}(S)m(S) = \gamma m(S_0) = M$, i.e. $\mathcal{A} \subseteq \arg\max_\Omega \varpi_{\tilde{w}} m$. S_0 is $\prec$-least in $\arg\max_\Omega \varpi_{\tilde{w}} m$, hence in its subset $\mathcal{A} \ni S_0$; that is, $S^*_{s_0} = S_0$. The other two formulations are restatements (contrapositive, and the event $\{S^*_{s_0} = S_0\}$ does not involve $\tilde{w}$). □

Remark 6.3 (the ceiling is approached). Assume $s_0 \le K$ and let $\tilde{w}_\epsilon(k) = \mathbf{1}\{k = s_0\} + \epsilon \mathbf{1}\{k \neq s_0\}$, $0 < \epsilon < 1$. On $\{S^*_{s_0} = S_0\} \cap \{m(S_0) > \epsilon \max_\Omega m\}$, every $S \in \Omega$ with $|S| \neq s_0$ has $\varpi_{\tilde{w}_\epsilon}(S)m(S) = \epsilon m(S)/\tilde{W} \le \epsilon \max_\Omega m/\tilde{W} < m(S_0)/\tilde{W} = \varpi_{\tilde{w}_\epsilon}(S_0)m(S_0)$, so $\arg\max_\Omega \varpi_{\tilde{w}_\epsilon} m = \arg\max_{|S|=s_0} m$, whose $\prec$-least element is S_0; thus $S^\dagger_{\tilde{w}_\epsilon} = S_0$. As $\epsilon \downarrow 0$ these events increase to $\{S^*_{s_0} = S_0\}$ (because $m(S_0) > 0$), so for any law of the data, $\sup_{\tilde{w}} \Pr(S^\dagger_{\tilde{w}} = S_0) = \Pr(S^*_{s_0} = S_0)$ by Theorem 6.2 and continuity from below. The maximising sequence depends on s_0, i.e. on the unknown truth.

For the decision results, let $q = (q_1, \dots, q_p) \in [0,1]^p$ be any vector and put $\psi_q(a) = \sum_{j \in a}(1 - 2q_j)$ for $a \subseteq [p]$, $J_+(q) = \{j : q_j > \frac{1}{2}\}$.

Lemma 6.4 (minimising ψ_q)**.** *(a) $J_+(q)$ minimises ψ_q over all subsets of $[p]$; it is the unique minimiser iff no $q_j = \frac{1}{2}$. (b) Let $0 \le K' \le p$, $\ell = \min(K', |J_+(q)|)$, and let $a^* \subseteq J_+(q)$ with $|a^*| = \ell$ and $\min_{j \in a^*} q_j \ge \max_{j \in J_+(q) \setminus a^*} q_j$ (conventions $\min \emptyset = +\infty$, $\max \emptyset = -\infty$; such a^* exist, and any deterministic tie-breaking rule at the boundary selects one). Then a^* minimises ψ_q over $\{a : |a| \le K'\}$.*

Proof. Write $c_j = 1 - 2q_j$: $c_j < 0$ on $J_+ = J_+(q)$, $c_j = 0$ if $q_j = \frac{1}{2}$, $c_j > 0$ otherwise, and c_j is strictly decreasing in q_j. (a) $\psi_q(a) = \sum_{j \in a} c_j \ge \sum_{j \in a, c_j < 0} c_j \ge \sum_{j : c_j < 0} c_j = \psi_q(J_+)$. Equality throughout forces $a \supseteq J_+$ and $a \setminus J_+ \subseteq \{c_j = 0\}$; so the minimisers are exactly $J_+ \cup E$ with $E \subseteq \{j : q_j = \frac{1}{2}\}$, unique iff that set is empty. (b) Existence of a^*: order J_+ by decreasing q_j (ties in any fixed way) and take the first ℓ. Let $|a| \le K'$ and $a' = a \cap J_+$. Then $\psi_q(a) \ge \psi_q(a')$ (removed terms have $c_j \ge 0$), and $|a'| \le \min(K', |J_+|) = \ell$. The values $\{c_j\}_{j \in a^*}$ are the ℓ smallest of $\{c_j\}_{j \in J_+}$ (with multiplicity): by the defining inequality and monotonicity of c in q, every c_j, $j \in J_+ \setminus a^*$, is $\ge$ every c_k, $k \in a^*$. The sum of any $r = |a'| \le \ell$ elements of $\{c_j\}_{j \in J_+}$ is at least the sum of the r smallest, which is at least the sum of the ℓ smallest because the further $\ell - r$ terms are negative. Hence $\psi_q(a') \ge \psi_q(a^*)$, and $|a^*| = \ell \le K'$. □

Theorem 6.5 (T07(ii), Bayes decisions)**.** *Let $\rho_L(a) = \sum_{S \in \Omega} \pi(S) L(a, S)$. (a) For $L(a, S) = \mathbf{1}\{a \neq S\}$, $a \subseteq [p]$: $\rho(a) = 1 - \pi(a)$. The minimisers over all $a \subseteq [p]$ are exactly the elements of $\arg\max_\Omega F$; in particular $S^\dagger$ is a minimiser, and the unique one iff $\arg\max_\Omega F$ is a singleton. (b) For $L(a, S) = |a \triangle S|$, $a \subseteq [p]$: $\rho_H(a) = \sum_j \mathrm{PIP}_j + \psi_{\mathrm{PIP}}(a)$. MPM minimises ρ_H over all subsets of $[p]$, uniquely iff no $\mathrm{PIP}_j = \frac{1}{2}$. (c) With $J_+ = \mathrm{MPM}$, $N_+ = |J_+|$, $\ell = \min(K, N_+)$ and $a^* \subseteq J_+$ the ℓ elements with the largest PIPs (boundary ties broken by a fixed deterministic rule), a^* minimises ρ_H over Ω.*

Proof. (a) $\rho(a) = \sum_S \pi(S) \mathbf{1}\{S \neq a\} = 1 - \pi(a)$. Minimising it means maximising $\pi(a)$ over $2^{[p]}$; $\pi > 0$ on $\Omega \neq \emptyset$ and $\pi = 0$ off Ω, so the maximisers lie in Ω, where $\pi = e^F/Z$; they are $\arg\max_\Omega F$, which contains $S^\dagger$ by definition. (b) $|a \triangle S| = \sum_j [\mathbf{1}\{j \in a, j \notin S\} + \mathbf{1}\{j \notin a, j \in S\}]$; taking

π-expectations, $\rho_H(a) = \sum_{j\in a}(1 - \mathrm{PIP}_j) + \sum_{j\notin a} \mathrm{PIP}_j = \sum_j \mathrm{PIP}_j + \sum_{j\in a}(1 - 2\mathrm{PIP}_j)$. Apply Lemma 6.4(a) with $q = \mathrm{PIP}$. (c) By (b), minimising ρ_H over $\Omega = \{a : |a| \le K\}$ is minimising ψ_{PIP} there; apply Lemma 6.4(b) with $q = \mathrm{PIP}$, $K' = K$. □

Remark 6.6. Since π is supported on Ω, $\sum_j \mathrm{PIP}_j = \mathbb{E}_\pi|S| \le K$; if $N_+ \ge 1$, $\sum_{j\in J_+} \mathrm{PIP}_j > N_+/2$, so $N_+ < 2K$. The cap therefore binds (i.e. $a^* \ne \mathrm{MPM}$) only when $K < N_+ < 2K$.

Theorem 6.7 (T07(iii), target versus reported decisions)**.** *Let $\mathcal{S} \subseteq \Omega$ be nonempty and $\hat{S}$ the $\prec$-least element of $\arg\max_{\mathcal{S}} F$. (a) $\hat{S} = S^\dagger$ iff $S^\dagger \in \mathcal{S}$. $\hat{S}$ minimises the posterior 0–1 loss iff $\mathcal{S} \cap \arg\max_\Omega F \ne \emptyset$. (b) $\{\hat{S} = S_0\} \cap \{S^*_{s_0} \ne S_0\} \subseteq \{S^*_{s_0} \notin \mathcal{S}\}$. (c) Let $\hat{p} \in [0,1]^p$, let $\mathcal{A}$ be either $2^{[p]}$ or Ω, and let $\hat{a}$ minimise $\psi_{\hat{p}}$ over $\mathcal{A}$ (e.g. $\widehat{\mathrm{MPM}} = J_+(\hat{p})$ for $\mathcal{A} = 2^{[p]}$, or the rule of Theorem 6.5(c) applied to $\hat{p}$ for $\mathcal{A} = \Omega$; both are minimisers by Lemma 6.4). For every minimiser a_{opt} of ρ_H over $\mathcal{A}$,*

$$0 \le \rho_H(\hat{a}) - \rho_H(a_{\mathrm{opt}}) \le 2 \sum_{j\in \hat{a}\triangle a_{\mathrm{opt}}} |\hat{p}_j - \mathrm{PIP}_j| \le 2\|\hat{p} - \mathrm{PIP}\|_1.$$

Also $\widehat{\mathrm{MPM}} = \mathrm{MPM}$ iff, for every j, $\hat{p}_j > \frac{1}{2} \Leftrightarrow \mathrm{PIP}_j > \frac{1}{2}$.

Proof. (a) If $S^\dagger \in \mathcal{S}$: $\max_{\mathcal{S}} F \le \max_\Omega F = F(S^\dagger) \le \max_{\mathcal{S}} F$, so $\max_{\mathcal{S}} F = \max_\Omega F$ and $\arg\max_{\mathcal{S}} F = \mathcal{S} \cap \arg\max_\Omega F \ni S^\dagger$. $S^\dagger$ is $\prec$-least in $\arg\max_\Omega F$, hence in this subset, so $\hat{S} = S^\dagger$. Conversely, $\hat{S} \in \mathcal{S}$, so $\hat{S} = S^\dagger$ forces $S^\dagger \in \mathcal{S}$. For the second claim, by Theorem 6.5(a), $\hat{S}$ minimises the 0–1 risk iff $\hat{S} \in \arg\max_\Omega F$. If $\mathcal{S} \cap \arg\max_\Omega F \ne \emptyset$ then, as above, $\max_{\mathcal{S}} F = \max_\Omega F$ and $\hat{S} \in \arg\max_{\mathcal{S}} F \subseteq \arg\max_\Omega F$. Conversely $\hat{S} \in \mathcal{S}$. (b) Suppose $\hat{S} = S_0$, $S^* := S^*_{s_0} \ne S_0$ and, for contradiction, $S^* \in \mathcal{S}$. Since $S_0 = \hat{S} \in \mathcal{S} \subseteq \Omega$ and $|S^*| = s_0$, both lie in Ω with the same prior weight, so $F(S^*) - F(S_0) = \log m(S^*) - \log m(S_0) \ge 0$ by the definition of S^*. If > 0, $S_0 \notin \arg\max_{\mathcal{S}} F$, contradicting $\hat{S} = S_0$. If $= 0$, then $S_0 \in \arg\max_{|S|=s_0} m$ and S^*, the $\prec$-least element there, satisfies $S^* \prec S_0$; both are in $\arg\max_{\mathcal{S}} F$, so $\hat{S} \ne S_0$, a contradiction. (c) Write $\psi = \psi_{\mathrm{PIP}}$ and $\hat{\psi} = \psi_{\hat{p}}$, so $\rho_H = \sum_j \mathrm{PIP}_j + \psi$ by Theorem 6.5(b). The lower bound holds because $\hat{a} \in \mathcal{A}$ and a_{opt} minimises over $\mathcal{A}$. For the upper bound,

$$\psi(\hat{a}) - \psi(a_{\mathrm{opt}}) = [\psi(\hat{a}) - \hat{\psi}(\hat{a})] + [\hat{\psi}(\hat{a}) - \hat{\psi}(a_{\mathrm{opt}})] + [\hat{\psi}(a_{\mathrm{opt}}) - \psi(a_{\mathrm{opt}})],$$

where the middle bracket is ≤ 0. With $c_j - \hat{c}_j = 2(\hat{p}_j - \mathrm{PIP}_j)$, the outer brackets sum to $\sum_{j\in\hat{a}}(c_j - \hat{c}_j) - \sum_{j\in a_{\mathrm{opt}}}(c_j - \hat{c}_j) = \sum_{j\in\hat{a}\setminus a_{\mathrm{opt}}}(c_j - \hat{c}_j) - \sum_{j\in a_{\mathrm{opt}}\setminus\hat{a}}(c_j - \hat{c}_j) \le 2\sum_{j\in\hat{a}\triangle a_{\mathrm{opt}}} |\hat{p}_j - \mathrm{PIP}_j|$. The last claim is the definition of $\widehat{\mathrm{MPM}} = J_+(\hat{p})$ and $\mathrm{MPM} = J_+(\mathrm{PIP})$. □

Remark 6.8. Theorem 6.7 identifies exactly what the guarantees of Theorems 6.2–6.5 need in order to transfer to the reported objects: the scoring events $\{S^\dagger \in \mathcal{S}\}$ or $\{\mathcal{S} \cap \arg\max_\Omega F \ne \emptyset\}$, and control of $\hat{p} - \mathrm{PIP}$. Neither is implied by (i)–(ii), which concern the target only: for instance, with $\mathcal{S} = \{S_0\}$ (when $S_0 \in \Omega$) one has $\hat{S} = S_0$ for every response, including on $\{S^*_{s_0} \ne S_0\}$, where $S^\dagger \ne S_0$ by Theorem 6.2; and with $\mathcal{S} = \{\emptyset\}$, $\hat{S} = \emptyset$ whatever $S^\dagger$ is. Bounds on them are computational results (T21, T23, T26), which this document neither uses nor proves.